WORKSHEET

Chapter 6: Real and imaginary factors

1 Solve $z^3 + (1 - 5i)z^2 + (-5 + 2i)z + 15 - 3i = 0$.

2 Solve $z^3 + (8 - i)z^2 + (19 + 3i)z + 26 + 26i = 0$.

3 Solve $z^3 + (3 + 2i)z^2 + (-9 + 10i)z - 23 + 24i = 0$.

4 Solve $z^3 + 5iz^2 + (-1 + 16i)z - 71 + 17i = 0$.

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5 Solve $z^4 + (2 + 2i)z^3 + (13 + 12i)z^2 + (26 + 7i)z + 30 + 84i = 0$.

6 Solve $6z^4 + (6 - 13i)z^3 + (6 + 29i)z^2 + (7 - 5i)z + 2 + 10i = 0$.

7 Solve $z^4 + (-1 + 7i)z^3 + (11 - 10i)z^2 + (-41 + 267i)z - 810 - 330i = 0$.

8 Solve $3z^4 + (24 - 34i)z^3 - (81 + 164i)z^2 - (216 + 34i)z - 24 - 224i = 0$.

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Answers

1 $z = 3i, 1 - i, -2 + 3i$

2 $z = -2i, -5 + i, -3 + 2i$

3 $z = -2 + i, 3 - 2i, -4 - i$

4 $z = -1 + 3i, 2 - \frac{5}{2}i, 3 - 2i$

5 $z = 3i, -2i, 1 - 4i, -3 + 2i$

6 $z = 6i, -5i, 4 - 3i, -3 - 5i$

7 $z = -\frac{1}{2}i, \frac{2}{3}i, 1 - i, -2 + 3i$

8 $z = \frac{2}{3}i, 4i, -2 + 3i, -6 + 5i$

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